Business Statistics

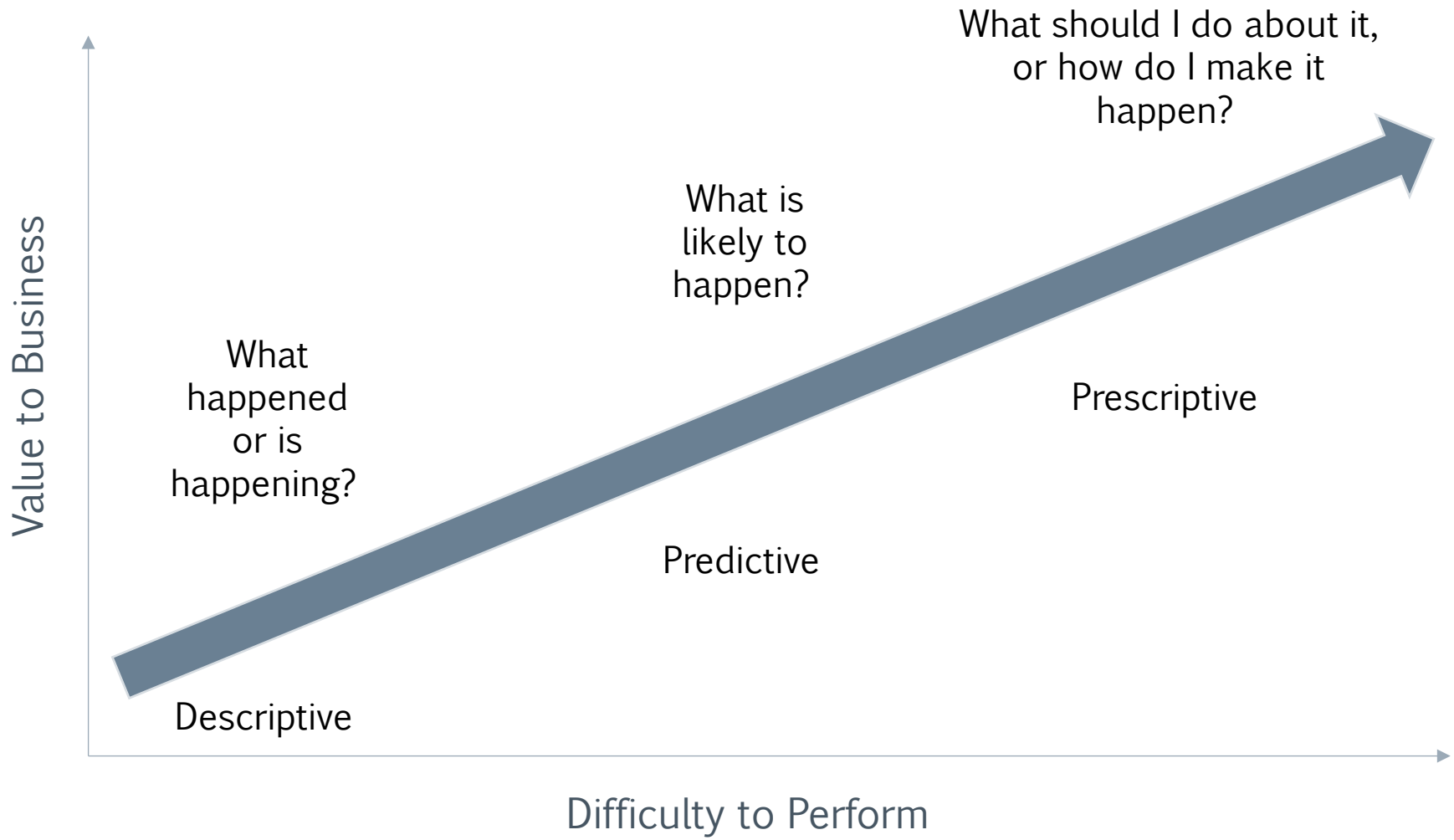
Why study statistics?

Datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

Statistics
Business decision making

- Business managers must constantly make decisions
- Data-driven decisions are becoming the norm
- Even managers who leverage their experience rely on data to quantify and understand the context, options, and risks
- Anyone promoting or advocating for a particular course of action does well to have data to back up their recommendations

Business Analytics
Value to the business



Business Decisions
Start with questions

- Decision: Should we improve our customers' experience?
- Question: How satisfied are our customers with our customer service?

- Decision: Should we change our employee remote-work policy?
- Question: Does remote work affect employee productivity?

- Decision: Should we improve the reliability of our product?
- Question: What proportion of units produced are defective?

- Decision: Should we improve the quality of high-school education?
- Question: How does education quality vary across public high schools in the United States

Statistics
Turning data into information

- There are massive amounts of data:
  - Facts, observations, quantitative and not
- By themselves data lack collective meaning
- Data alone is not information
- Information is meaningful and useful
- With statistics we can transform data into useful information to answer questions and make decisions

age	weight	sex	profession
22	150	F	Student
35	185	M	Accountant
41	160	F	Nurse
29	172	M	IT Tech
54	190	M	Manager

Statistics
What is statistics?

- There will be more technical definitions later.
- At this point, lets just consider statistics as the use of quantitative tools and techniques to learn about the data, develop information, test our guesses, and make predictions.
- In this course we will learn about:
  - Data types and their visualization
  - Finding a central value of the data and how its values are spread out
  - Probability and how it is distributed for different uncertain values
  - How to choose and use samples
  - Estimating values with intervals
  - Testing our beliefs about the data
  - Making comparisons between different sets or subsets of data
  - Estimating the relationship between variables to explain the data and make predictions
  - Make forecasts over time